# Complexity of the cluster deletion problem on chordal graphs, subclasses of chordal graphs, and cographs*

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**Abstract**

We consider the following vertex-partition problem on graphs: given a graph with real nonnegative edge weights, partition the vertices into clusters (i.e. cliques) to minimize the total weight of edges out of the clusters. This optimization problem is known to be an NP-complete problem even for unweighted graphs and has been studied extensively in the scope of fixed-parameter tractability (FPT), where it is commonly known as the CLUSTER DELETION problem. Many of the recently-developed FPT algorithms rely on being able to solve CLUSTER DELETION in polynomial-time on restricted graph structures. In this paper, the complexity of the CLUSTER DELETION problem is investigated for the family of chordal graphs. It is shown that this problem is NP-complete for edge-weighted split graphs, edge-weighted interval graphs and edge-unweighted chordal graphs. Some polynomial-time solvable cases are also identified, in particular CLUSTER DELETION for unweighted split graphs, unweighted proper interval graphs and weighted block graphs.

**Keywords**: Block graphs, Cliques, Edge-deletion, Cluster deletion, Interval graphs, Split graphs, Submodular functions, Chordal graphs, NP-completeness.

## 1 Introduction

A *cluster graph* is a graph in which every connected component is a clique (i.e. a complete subgraph). Cluster graphs have been used in a variety of applications whenever clustering of objects is studied or when consistent data is sought among noisy or error-prone data [1, 4]. The *cluster deletion* problem asks for the minimum number of edges that can be removed from an input graph to make the resulting graph a cluster graph. There exist several results for the cluster deletion problem (see for example [3, 15, 12] and references therein). The cluster deletion problem is known to be NP-complete [15] for general graphs. Moreover, Shamir et al. [15] showed that it remains NP-hard when enforcing that the input graph should be clustered into exactly $d \geq 3$ components. They also showed that when the input graph should be clustered into exactly 2 components, the

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problem is polynomial-time solvable. Komusiewicz et al. [12] proved that cluster deletion is hard for $C_4$-free graphs with maximum degree 4, and they gave a $O(n^{1.5}\log^2 n)$ time algorithm for solving cluster deletion on graphs with maximum degree 3. Based on results obtained by Demaine et al. [5] concerning a variant of a clustering problem, Dessmark et al. [6] provided a polynomial $O(\log n)$-approximation algorithm for the edge-weighted version of the cluster deletion problem. They also showed that for the edge-unweighted version of cluster deletion on general graphs, the greedy algorithm that finds iteratively maximum cliques gives a 2-approximation algorithm to the optimal cluster deletion. The complexity of such an algorithm relies on the complexity of iteratively finding maximum cliques, so it is a polynomial-time approximation algorithm for certain graph classes. Recently, Gao et al. [9] showed that the greedy algorithm that finds iteratively maximum cliques, gives an optimal solution for the class of graphs known as *cographs*. It implies that the cluster deletion problem is polynomial-time solvable on unweighted cographs. Moreover, Gao et al. [9] showed that the cluster deletion problem is NP-hard on $(C_5, P_5)$-free graphs, on $(2K_2, 3K_1)$-free graphs and on $(C_5, P_5, \text{bull}, 4\text{-pan}, \text{fork}, \text{co-gem}, \text{co-4-pan})$-free graphs. For weighted graphs, the cluster deletion problem can be solved in polynomial time on the class of triangle-free graphs, since it is equivalent to maximum weighted matching [7]. The cluster deletion problem has been studied extensively in the scope of fixed-parameter tractability (FPT). Many of the recently-developed FPT algorithms rely on being able to solve cluster deletion in polynomial-time on restricted graph structures [3].

This paper is organized as follows. In Section 2, we prove that the weighted cluster deletion problem is NP-complete in a subclass of split graphs and even under certain restrictions for the weights, and that it is polynomial-time solvable for unweighted split graphs. In Section 3, we prove that the unweighted cluster deletion problem is NP-complete in a subclass of chordal graphs, namely $P_5$-free chordal graphs. In Section 4, we prove that the weighted cluster deletion problem is polynomial-time solvable on block graphs, when the weights are nonnegative. In Section 5, we show that the weighted cluster deletion problem is NP-complete on interval graphs and, moreover, on complete split graphs. On the positive side, we show that the unweighted problem is polynomial-time solvable for proper interval graphs, and that the weighted cluster deletion problem is polynomial-time solvable on a subclass of proper interval graphs known as paths of cliques. As a corollary of the complexity on complete split graphs, it is shown that the weighted cluster deletion problem is NP-complete on cographs, in contrast to the polynomial-time algorithm given in [9] for the unweighted case.

We conclude this introduction with some definitions. Let $G = (V, E)$ be a graph. For each vertex $v \in V$, we denote by $N(v) = \{u : vu \in E\}$ the set of neighbors of $v$ in $G$. Two vertices $v$ and $w$ are called *true twins* if $N(v) \cup \{v\} = N(w) \cup \{w\}$. A graph $G$ is called *weighted* if there is a weight function $w : E \to \mathbb{R}^+$ associated with it. For the algorithms involving weighted graphs, we will assume that the weights are rational (or belong to any ordered field in which we can perform the operations algorithmically). An *unweighted* graph is a graph in which each edge has a weight equal to 1. A graph is *chordal*, if and only if it does not contain a cycle of length at least four as an induced subgraph. Given a vertex partition $S = C_1, \ldots, C_k$ of a graph $G$, we call the *weight* of $S$ to the sum of the weight of all edges $e = uv$ such that $u \in C_i$, $v \in C_j$, with $i \neq j$. The cluster deletion problem of a (un)weighted graph $G$ can be redefined as the problem of finding a clique partition of $G$ with minimum weight.

## 2 Split graphs

A graph $G = (V, E)$ is a *split graph*, if and only if there is a partition of the vertices $V$ of $G$ into a clique $K$ and an independent set $I$. Another necessary and sufficient condition for a graph $G$ to be a split graph is that $G$ and its complement $\overline{G}$ are chordal graphs (see [8]).

In this section we prove the NP-completeness of the weighted cluster deletion problem for split graphs by reduction from the *exact cover by 3-sets* problem (*X3C* problem for short). The formal definition of the X3C problem can be stated as follows:

**Instance:** A set $X$ with $3q$ elements and a collection $C$ of 3-element subsets of $X$.

**Question:** Does $C$ contain an exact cover for $X$? In other words, is there a subset $C' \subseteq C$, such that every element of $X$ occurs in exactly one member of $C'$?

The X3C problem is known to be NP-complete [10].

We analyze in this section a subclass of the split graphs, namely the class of those split graphs where each vertex of the independent set $I$ is adjacent to exactly $p$ vertices of the clique $K$. We call these graphs the $p$-split graphs, where $p \geq 1$.

**Theorem 1** (NP-completeness on weighted 3-split graphs)**.** *The cluster deletion problem is NP-complete for weighted* 3*-split graphs, even if the weight of all the internal edges of the clique is* 1 *and the weight of the edges between the clique and the stable set is uniform.*

*Proof.* It is not hard to see that the cluster deletion problem is in NP, since we can easily verify in polynomial time whether a vertex partition of a graph is a clique partition or not, and if it has weight less that a given value $k$. Let $X = \{x_1, \ldots, x_{3q}\}$ and $C = \{c_1, \ldots, c_m\}$ be an instance of the X3C problem, where each element $c_i \in C$ is a 3-element subset of $X$, with $m \geq q \geq 1$. We want to know if there exists a subset $C' \subseteq C$ with size $q$ such that each element in $X$ belongs to exactly one of the elements in $C'$. We will construct an edge-weighted split graph $G = (K_X \cup I_C, E)$, where $K_X$ induces a clique with $3q$ vertices and $I_C$ induces an independent set with $m$ vertices. In fact, for each element $x_i \in X$ there is a vertex $v_{x_i}$ in $K_X$ and for each 3-subset $c_j \in C$ there is a vertex $v_{c_j} \in I_C$. The edge set $E$ is formed as follows: $E = E_X \cup E_C$, where $E_X = \{v_{x_i}v_{x_j} : i \neq j, v_{x_i}, v_{x_j} \in K_X\}$ and $E_C = \{v_{c_j}v_{x_{j_1}}, v_{c_j}v_{x_{j_2}}, v_{c_j}v_{x_{j_3}} : v_{c_j} \in I_C, v_{x_{j_1}}, v_{x_{j_2}}, v_{x_{j_3}} \in K_X, c_j \in C \text{ and } c_j = \{x_{j_1}, x_{j_2}, x_{j_3}\}\}$. Finally, the weight of each edge in $E_X$ is equal to 1 and the weight of each edge in $E_C$ is equal to $\beta$, where $\beta = 3q$. Clearly, $G$ is an split graph and its construction is done in polynomial time. Let $W = \binom{3q}{2} - 3q + 3(m-q)\beta$. We will show that there exists a subset $C' \subseteq C$, with $|C'| = q$, exactly covering $X$ if and only if $G$ has a clique partition where the sum of the weights of edges outside the cliques is at most $W$. In other words, there exists a solution for the X3C problem if and only if there exists an optimal solution for the cluster deletion problem for $G$ with weight at most $W$.

Assume first that there exists $C' \subseteq C$, with $C' = \{c'_1, \ldots, c'_q\}$ such that $c'_i \cap c'_j = \emptyset$ whenever $i \neq j$, and $\bigcup_{c'_j \in C'} c'_j = X$. The clique partition for $G$ can be constructed as follows: for each $c'_j \in C'$, with $c'_j = \{x_{j_1}, x_{j_2}, x_{j_3}\}$, choose the clique $\{v_{c'_j}, v_{x_{j_1}}, v_{x_{j_2}}, v_{x_{j_3}}\}$ in $G$. Each one of the remaining $m - q$ vertices in $I_C$ form a clique with size one. It is easy to see that the sum of the edge weights outside those cliques is exactly equal to $W$.

Conversely, assume that $G$ has a clique partition with weight at most $W$. We should to prove

that there exists $C' \subseteq C$, with $|C'| = q$, such that $C'$ is an exact cover for $X$. For this, we analyze first the structure of the optimal solutions of the cluster deletion problem for $G$.

**Claim 1.** *Let $S$ be a clique partition of $G$ which is an optimal solution for the cluster deletion problem in $G$. If there exists a clique in $S$ formed only by vertices in $K_X$ then, there is no solution to the X3C problem.*

First, notice that it is not possible to have two cliques, say $A_1, A_2 \in S$, formed only by vertices in $K_X$. Otherwise, set $S' = S \setminus \{A_1, A_2\} \cup \{A_1 \cup A_2\}$. Clearly, the weight of $S'$ is $|A_1|.|A_2| \geq 1$ less than the weight of $S$, which is a contradiction to the optimally of $S$. On the other hand, every clique of $S$ contains at most one vertex of $I$, so there are $m$ cliques (possibly singletons) containing one vertex of $I$ each. Let $S = A_1, B_1, \ldots, B_m$ be the clique partition of $G$ and let $W_S$ be the weight of $S$. Assume that the clique $A_1$ is formed only by vertices in $K_X$. Clearly, each clique $B_j$, $1 \leq j \leq m$, contains a vertex in $I_C$ and zero, one, two or three vertices in $K_X$. We consider the following cases:

- Suppose there exists $B_j = \{v_{c_j}, v_{x_{j_1}}, v_{x_{j_2}}\}$ in $S$ such that there is $v_y \in A_1$ with $v_y$ be adjacent to $v_{c_j}$, where $v_{c_j} \in I_C$ and $v_y, v_{x_{j_1}}, v_{x_{j_2}} \in K_X$. Let $S'$ be the clique partition obtained by setting $B'_j \leftarrow B_j \cup \{v_y\}$ and $A'_1 \leftarrow A_1 \setminus \{v_y\}$, where $S' = S \setminus \{A_1, B_j\} \cup \{A'_1, B'_j\}$. Let $W_{S'}$ be the weight of $S'$. It is not difficult to see that $W_{S'} = W_S - (\beta + 2) + (|A_1| - 1) = W_S + |A_1| - \beta - 3$. However, $\beta = 3q$ and $|A_1| \leq 3q - 2$. Therefore, $W_{S'} < W_S$, which contradicts the optimality of $S$.

- Similarly to the previous case, suppose there exists $B_j = \{v_{c_j}, v_{x_{j_1}}\}$ (resp. $B_j = \{v_{c_j}\}$) in $S$ such that there is $v_y \in A_1$ with $v_y$ be adjacent to $v_{c_j}$, where $v_{c_j} \in I_C$ and $v_y, v_{x_{j_1}} \in K_X$ (resp. $v_y \in K_X$) . By setting $B'_j \leftarrow B_j \cup \{v_y\}$ and $A'_1 \leftarrow A_1 \setminus \{v_y\}$, we obtain a new clique partition $S' = S \setminus \{A_1, B_j\} \cup \{A'_1, B'_j\}$ which weight $W_{S'} = W_S + |A_1| - \beta - 2 < W_S$ (resp. $W_{S'} = W_S + |A_1| - \beta - 1 < W_S$), which is a contradiction to the optimality of $S$.

As $S$ is an optimal solution, no vertex in $I_C$ is adjacent to any vertex in $A_1$, but by construction, it implies that $X$ is not covered by $C$, and thus, there is no solution for the X3C problem. This ends the proof of this claim. $\diamondsuit$

By the previous claim, if $\bigcup_{c_j \in C} c_j = X$ then, an optimal solution $S$ for the cluster deletion problem of $G$ must be of the form $B_1, \ldots, B_m$, where each clique $B_j$ contains exactly one vertex $v_{c_j} \in I_C$, for $1 \leq j \leq m$. Now, we can deduce the following result on $S$.

**Claim 2.** *Let $S = B_1, \ldots, B_m$ be an optimal solution of the cluster deletion problem for $G$. Let $m = t_1 + t_2 + t_3 + t_4$, with $t_i \geq 0$, for $1 \leq i \leq 4$, where $t_1$ (resp. $t_2$, $t_3$, $t_4$) represents the number of cliques $B_j$ in $S$ formed by a vertex in $I_C$ and by three (resp. two, one, zero) vertices in $K_X$. Then, (i) $t_1 > 0$; (ii) Let $v_{c_j}$ be a vertex in $I_C$ such that it belongs to a clique $B_j$ of $S$ with $|B_j| = k$, with $1 < k \leq 3$. Then, for each vertex $x \in N(v_{c_j})$ such that $x \notin B_j$, we have that $x$ must belong to another clique $B_i$ of $S$ with $|B_i| > k$.*

Suppose that $t_1 = 0$. If $t_2 \neq 0$ then, there is a clique $B_j = \{v_{c_j}, v_{x_{j_1}}, v_{x_{j_2}}\}$ in $S$. By construction, $v_{c_j}$ has another neighbor $v_y \in K_X$. If $v_y \in B_i$, with $|B_i| = 3$ (resp. $|B_i| = 2$) then, we can construct another clique partition $S'$ by setting $B'_j \leftarrow B_j \cup \{v_y\}$ and $B'_i \leftarrow B_i \setminus \{v_y\}$ and where $S' = (S \setminus \{B_i, B_j\}) \cup \{B'_j \cup B'_i\}$. Notice that the weight $W_{S'}$ of $S'$ is equal to $W_S - (\beta + 2) + \beta + 1 = W_S - 1 < W_S$ (resp. $W_S - (\beta + 2) + \beta = W_S - 2 < W_S$) which contradicts the optimality of $S$. If $t_1 = t_2 = 0$ then, there are $B_j, B_i, B_s \in S$ such that $B_j = \{v_{c_j}, v_{x_j}\}$, $B_i = \{v_{c_i}, v_{x_i}\}$ and $B_s = \{v_{c_s}, v_{x_s}\}$ where

$\{v_{x_j}, v_{x_i}, v_{x_s}\}$ are the three neighbors of $v_{c_j}$. By setting $B'_j \leftarrow B_j \cup \{v_{x_i}, v_{x_j}\}$, $B'_i \leftarrow B_i \setminus \{v_{x_i}\}$, $B'_s \leftarrow B_s \setminus \{v_{x_s}\}$, and by constructing the partition $S' = S \setminus \{B_j, B_i, B_s\} \cup \{B'_j, B'_i, B'_s\}$ we have that the weight $W_{S'}$ of $S'$ is equal to $W_S - (2\beta + 3) + 2\beta = W_S - 3 < W_S$, which is again a contradiction to the optimality of $S$. Therefore, $t_1 > 0$ and thus $(i)$ holds. In an analogous way, we can show $(ii)$, which ends the proof of this claim. $\diamondsuit$

Now, let $S = B_1, \ldots, B_m$ be an optimal clique partition of $G$ with weight $W_S$. By Claim 1 we know that each clique $B_i$ in $S$ contains exactly one vertex in $I_C$ and that there are $t_1$ (res. $t_2$, $t_3$, $t_4$) cliques in $S$ having exactly three (resp. two, one, zero) vertices in $K_X$. By Claim 2, we know that $t_1 > 0$ and $t_2, t_3, t_4 \geq 0$. Therefore, the weight $W_S$ of $S$ can be written as: $W_S = \binom{3q}{2} - (3t_1 + t_2) + (t_2 + 2t_3 + 3t_4)\beta$. Moreover, $m = t_1 + t_2 + t_3 + t_4$ and $3q = 3t_1 + 2t_2 + t_3$. It is not difficult to see that $3(m - q) = t_2 + 2t_3 + 3t_4$ and that $3t_1 + t_2 = 3q - t_2 - t3$ which implies that $3q \geq 3t_1 + t_2$. Therefore, $W \leq W_S$ with equality only if $t_2 = t_3 = 0$ in which case, $t_1 = q$. Therefore, if $S$ has weight $W$ then, there is a solution to the X3C problem which ends the proof of this theorem. $\square$

For the unweighted case, we will show next that the problem can be easily solved on split graphs.

**Theorem 2** (Polynomiality on unweighted split graphs)**.** *The cluster deletion problem is polynomial-time solvable for unweighted split graphs. Indeed, if $(K, I)$ is a split partition of the graph $G$ such that $K$ is a maximal clique of $G$, then $\{K\} \cup \{\{v\} : v \in I\}$ is an optimal solution unless there is a vertex $v_1$ in $I$ adjacent to all but one vertex $w$ in $K$, and that vertex $w$ has a neighbor $v_2$ in $I$. In that case, an optimal solution is $\{\{v_1\} \cup (K - \{w\}), \{w, v_2\}\} \cup \{\{v\} : v \in I, v \neq v_1, v_2\}$.*

*Proof.* Let $G$ be a split graph and let $(K, I)$ be a split partition of $G$ such that $K$ is a maximal clique of $G$. We can see the problem as the problem of maximizing the internal edges in a clique partition $S$ (it is equivalent to the problem of minimizing the edges joining distinct cliques of $S$). In order to break ties, we will maximize the internal edges that join two vertices of $K$ in $G$. We will call the *standard partition* to $S_0 = \{K\} \cup \{\{v\} : v \in I\}$.

Suppose that we have an optimal solution $S$ with respect to the criterion that is strictly better than $S_0$. If $S$ contains a clique $A \subsetneq K$ such that $|A| = a \geq 1$ and a clique $B$ containing one vertex of $I$ and $b \geq 1$ vertices of $K$, then $S' = S \setminus \{A, B\} \cup \{A \cup (B \cap K), B \cap I\}$, with respect to $S$, has $ab$ new internal edges that join two vertices of $K$ in $G$ and looses $b$ edges that join vertices of $K$ with vertices of $I$, so it is either strictly better than $S$ or it is preferable by the breaking ties criterion, a contradiction to the optimality of $S$. So, we may assume that every clique of $S$ contains a vertex of $I$.

Suppose now that the clique $A$ has a vertex of $I$ and $a \geq 2$ vertices of $K$, and the clique $B$ has a vertex of $I$ and $b \geq 2$ vertices of $K$. Then $S' = S \setminus \{A, B\} \cup \{(A \cap K) \cup (B \cap K), A \cap I, B \cap I\}$, with respect to $S$, has $ab$ new internal edges that join two vertices of $K$ in $G$ and looses $a + b$ edges that join vertices of $K$ with vertices of $I$, so as $a$ and $b$ are at least 2, $S'$ is either strictly better than $S$ or it is preferable by the breaking ties criterion, a contradiction to the optimality of $S$.

Suppose now that there are three cliques $A, B, C$ having a vertex of $I$ each and $a, b, c \geq 1$ vertices of $K$, respectively. Then $S' = S \setminus \{A, B, C\} \cup \{(A \cap K) \cup (B \cap K) \cup (C \cap K), A \cap I, B \cap I, C \cap I\}$, with respect to $S$, has $ab + ac + bc$ new internal edges that join two vertices of $K$ in $G$ and looses $a + b + c$ edges that join vertices of $K$ with vertices of $I$, so as $ab \geq a$, $bc \geq b$, and $ca \geq c$, $S'$ is either strictly better than $S$ or it is preferable by the breaking ties criterion, a contradiction to the optimality of $S$.

Then $S$ has exactly two cliques containing vertices of $K$, each of them contains a vertex of $I$, and at least one of them has only one vertex of $K$. So, there is a vertex $v_1$ in $I$ adjacent to all but one vertex $w$ in $K$, and that vertex $w$ has a neighbor $v_2$ in $I$, and $S = \{\{v_1\} \cup (K - \{w\}), \{w, v_2\}\} \cup \{\{v\} : v \in I, v \neq v_1, v_2\}$. □

# 3 Chordal graphs

Chordal graphs form an extensively studied class of graphs, as their peculiar clique-based structure allows an efficient solution for algorithmic problems. A very nice survey by Blair and Peyton on chordal graphs properties can be found in [2].

In order to prove the main result of this section, we will show first a simple general lemma.

**Lemma 1** (True twins)**.** *Let $G$ be graph and $v$, $w$ be true twins in $G$. Then, for every optimal solution of the unweighted cluster deletion problem, $v$ and $w$ belong to the same clique of the partition.*

*Proof.* Suppose, on the contrary, that there is an optimal clique partition $S$ such that $v$ belongs to a clique $C_1$ and $w$ to a different clique $C_2$. Without loss of generality, we may assume $|C_1| \leq |C_2|$. But then $S' = S \setminus \{C_1, C_2\} \cup \{C_1 \setminus \{v\}, C_2 \cup \{v\}\}$ has strictly less weight than $S$, a contradiction. □

**Theorem 3** (NP-completeness on unweighted chordal graphs)**.** *The cluster deletion problem is NP-complete for unweighted $P_5$-free chordal graphs.*

*Proof.* The proof of this theorem is similar to the proof of Theorem 1. The reduction is done again from the X3C problem. Let $X = \{x_1, \ldots, x_{3q}\}$ and $C = \{c_1, \ldots, c_m\}$ be an instance of the X3C problem, where each element $c_i \in C$ is a 3-element subset of $X$, with $m \geq q \geq 1$. We want to know if there exists a subset $C' \subseteq C$ with size $q$ such that each element in $X$ belongs to exactly one of the elements in $C'$. We construct a graph $G = (V, E)$ as follows. The set of vertices $V$ is formed by the subsets $K_X, K_{c_1}, \ldots, K_{c_m}$, where $K_X = \{v_{x_i} : x_i \in X\}$ is the subset of vertices representing the elements of $X$. Let $\beta = 3q$. For each 3-subset $c_i \in C$, we construct a set of vertices $K_{c_i} = \{v_{c_{i,1}}, \ldots, v_{c_{i,\beta}}\}$ with cardinality $\beta$. The set of edges is constructed as follows: $K_X$ induces a clique of size $\beta$, and also each $K_{c_i}$ induces a clique of size $\beta$, for $1 \leq i \leq m$. Finally, for each $c_i \in C$ such that $c_i = \{x_{i_1}, x_{i_2}, x_{i_3}\}$, we add all the edges between the vertices of the clique $K_{c_i}$ to the vertices $v_{x_{i_1}}, v_{x_{i_2}}, v_{x_{i_3}} \in K_X$. Clearly, this construction is done in polynomial time and the obtained graph $G$ is an unweighted $P_5$-free chordal graph. Now, since for each $1 \leq i \leq m$, all the $\beta$ vertices of the clique $K_{c_i}$ are true twins, by Lemma 1 they must belong to the same clique in an optimal partition $S$. Therefore, we can contract each clique $K_{c_i}$, with $1 \leq i \leq m$, into a single vertex $c_i$ and replace each subset of $\beta$ edges between $K_{c_i}$ and the vertex $v_{x_{i_j}} \in K_X$ by a single edge with weight $\beta$, where $1 \leq j \leq 3$ and $x_{i_j} \in c_i$. This is exactly the weighted split graph used in Theorem 1. □

# 4 Block graphs

A graph $G$ is a *block* graph if it is a connected graph and every *block* (i.e. maximal 2-connected component) is a clique. Block graphs are a subclass of chordal graphs.

The first result in this section concerns weighted 1-split graphs, a particular subclass of block graphs. We will use submodular functions minimization in order to solve the cluster deletion

problem on 1-split graphs, and then we will reduce the problem on weighted block graphs to the problem on weighted 1-split graphs.

Given a finite nonempty set $V$ of cardinality $n$, a function $f$ defined on all the subsets of $V$ is called *submodular* if it satisfies $f(X) + f(Y) \geq f(X \cup Y) + f(X \cap Y)$, for every $X, Y \subseteq V$. In [11] and [14] are presented combinatorial polynomial-time algorithms for finding a minimizer of a general submodular function, provided that an oracle for evaluating the function value is available. The number of oracle calls is bounded by a polynomial in the size of the underlying set.

**Theorem 4** (Polynomiality on weighted 1-split graphs)**.** *Cluster deletion is polynomial time solvable for weighted* 1*-split graphs, provided that the weights of the internal edges of the clique are nonnegative.*

*Proof.* In the cluster deletion problem of a 1-split graph whose split partition is $(K, I)$, the solution has at most one clique $K_1$ that has no intersection with $I$, some pairs $\{v, u\}$ with $v \in K$ and $u \in I$, and some isolated vertices from $I$. Each vertex of $I$ has at most one neighbor in $K$, and if $\{v, u\}$ belongs to the solution, then $u$ is one of the neighbors of $v$ in $I$ maximizing the weight of the edge $vu$. Indeed, we can preprocess the graph, identifying a subset of vertices of $I$ that will be trivial cliques in the solution, and leaving a graph in which each vertex of $K$ has at most one neighbor in $I$. If the subset $K_0$ of vertices of $K$ having no neighbors in $I$ is nonempty, it will be part of the clique $K_1$ in the solution. We will name the set $K \setminus K_0$ as $\{v_1, \ldots, v_r\}$ and their respective neighbors in $I$ as $\{u_1, \ldots, u_r\}$. So, a candidate solution is totally determined by the subset $S$ of $\{1, \ldots, r\}$ such that the solution is $\{\{v_i, u_i\}\}_{i \notin S} \cup \{K_1 \cup \{v_i : i \in S\}\} \cup \{\{u_i\}\}_{i \in S}$. The subset $S$ is not necessarily a proper subset of $\{1, \ldots, r\}$ and it could be empty. We will prove that the function $f$ that assigns to $S$ the difference of weight between the candidate solution associated with $S$ and the solution associated with the empty set is submodular, and thus the set $S$ that minimizes that function can be found in polynomial time. The function $f$ can be computed in polynomial time for a subset $S$, and is defined as

$$f(S) = \sum_{i \in S} w(v_i u_i) - \sum_{i,j \in S} w(v_i v_j) - \sum_{i \in S, z \in K_0} w(v_i z)$$

Let $S, T \subseteq \{1, \ldots, r\}$. We will show that $f(S) + f(T) \geq f(S \cup T) + f(S \cap T)$, provided that the weights of the edges joining two vertices of $K$ are nonnegative. It is easy to see that $\sum_{i \in S} w(v_i u_i) + \sum_{i \in T} w(v_i u_i) = \sum_{i \in S \cup T} w(v_i u_i) + \sum_{i \in S \cap T} w(v_i u_i)$ and that $\sum_{i \in S, z \in K_0} w(v_i z) + \sum_{i \in T, z \in K_0} w(v_i z) = \sum_{i \in S \cup T, z \in K_0} w(v_i z) + \sum_{i \in S \cap T, z \in K_0} w(v_i z)$. So we have to show that $\sum_{i,j \in S} w(v_i v_j) + \sum_{i,j \in T} w(v_i v_j) \leq \sum_{i,j \in S \cup T} w(v_i v_j) + \sum_{i,j \in S \cap T} w(v_i v_j)$, but $\sum_{i,j \in S \cup T} w(v_i v_j) + \sum_{i,j \in S \cap T} w(v_i v_j) - \sum_{i,j \in S} w(v_i v_j) - \sum_{i,j \in T} w(v_i v_j) = \sum_{i \in S \setminus T, j \in T \setminus S} w(v_i v_j) \geq 0$.

If the weight of the internal edges is 1, the algorithm is very simple. We name the vertices in $K \setminus K_0$ as $\{v_1, \ldots, v_r\}$ and their respective neighbors in $I$ as $\{u_1, \ldots, u_r\}$, in such a way that $w(v_1 u_1) \leq \cdots \leq w(v_r u_r)$. Then the only sets that are candidates to minimize $f$ are the empty set and the sets $\{1, \ldots, j\}$, for $1 \leq j \leq r$. □

Based on the solution of cluster deletion for weighted 1-split graphs, we will solve the problem for weighted block graphs, when the weight of the edges is nonnegative.

**Theorem 5** (Polynomiality on weighted block graphs)**.** *Cluster deletion is polynomial time solvable for weighted block graphs, when the weight of the edges is nonnegative.*

*Proof.* Let $G = (V, E)$ be a weighted block graph, such that the weight $w$ of the edges is nonnegative. An *end-block* of a graph is a block containing exactly one cut-vertex. It is known that every connected graph that is not 2-connected has an end-block. Inductively, the blocks of $G$ can be enumerated as $B_1, \ldots, B_r$ in such a way that $B_i$ is an end-block of the graph induced by $B_i \cup B_{i+1} \cup \cdots \cup B_r$. We will process the blocks in that order by solving each time a subproblem and reducing iteratively the graph to a simpler graph. Then we will reconstruct the solution for the original graph based on the solution of each subproblem and the recursive solution of the reduced graph.

By the way of choosing the order of the blocks, when processing the block $B_i$ for $i < r$, it will have exactly one cut-vertex $v$ joining $B_i$ with non-processed blocks. The graph $G_0$ will be $G$, we will create a graph $G_i$ from $G_{i-1}$ by replacing the connected component of $G_{i-1} \setminus \{v\}$ that contains $B_i \setminus \{v\}$ by a vertex $u$ adjacent to $v$ and we will assign to the edge $vu$ a suitable weight. It means, inductively, that when processing block $B_i$, the connected component $H$ of $G_{i-1} \setminus \{v\}$ that contains $B_i \setminus \{v\}$ is a 1-split graph, and so is the graph induced by $V(H) \cup \{v\}$. We will define $w(uv) = \sum_{y \in H, vy \in E} w(vy) + w(S_H) - w(S_{H \cup \{v\}})$, where $S_{H \cup \{v\}}$ and $S_H$ are optimal solutions for the cluster deletion problem on the 1-split graph induced by $V(H) \cup \{v\}$, and $H$, respectively. Notice that we can find $S_{H \cup \{v\}}$ and $S_H$ by Theorem 4.

**Claim 3.** *Let $S_i$ be an optimal solution of the cluster deletion problem for $G_i$. If $\{u, v\} \in S_i$, then $S_i \setminus \{\{u, v\}\} \cup S_{H \cup \{v\}}$ is an optimal solution for $G_{i-1}$. If $\{u, v\} \notin S_i$, then $S_i \setminus \{\{u\}\} \cup S_H$ is an optimal solution for $G_{i-1}$.*

Let $S_{i-1}$ be an optimal solution of the cluster deletion problem for $G_{i-1}$. Let $S^1_{i-1}$ the subset of cliques of $S_{i-1}$ containing vertices of $H$ and $S^2_{i-1} = S_{i-1} \setminus S^1_{i-1}$. We have that either $v$ belongs to a clique in $S^1_{i-1}$ or not. As $v$ is a cut-vertex of $G_{i-1}$, it is clear that in the first case $S^1_{i-1}$ is an optimal solution for the cluster deletion problem on the graph induced by $V(H) \cup \{v\}$, while in the second case $S^1_{i-1}$ is an optimal solution for the cluster deletion problem on the graph $H$. On the other hand, in the first case, $S^2_{i-1} \cup \{v, u\}$ will be a feasible solution for $G_i$, with weight $w(S_{i-1}) - w(S_{H \cup \{v\}})$, while in the second case, $S^2_{i-1} \cup \{u\}$ will be a feasible solution for $G_i$, with weight $w(S_{i-1}) - w(S_H) - \sum_{y \in H, vy \in E} w(vy) + w(uv) = w(S_{i-1}) - w(S_{H \cup \{v\}})$. So, if $S_i$ is an optimal solution of the cluster deletion problem for $G_i$, then $w(S_i) \leq w(S_{i-1}) - w(S_{H \cup \{v\}})$, which implies $w(S_i) + w(S_{H \cup \{v\}}) \leq w(S_{i-1})$.

Now, let us see that the weight of the solutions proposed for $G_{i-1}$ in each case is exactly $w(S_i) + w(S_{H \cup \{v\}})$, which makes them respectively optimal.

If $\{u, v\} \in S_i$, it is easy to see that $w(S_i \setminus \{\{u, v\}\} \cup S_{H \cup \{v\}}) = w(S_i) + w(S_{H \cup \{v\}})$. If $\{u, v\} \notin S_i$, when considering $S_i \setminus \{\{u\}\} \cup S_H$ in $G_{i-1}$, we do not have to delete the edge $uv$, but we have to delete every edge joining $v$ with vertices in $H$, so $w(S_i \setminus \{\{u\}\} \cup S_H) = w(S_i) - w(uv) + \sum_{y \in H, vy \in E} w(vy) + w(S_H) = w(S_i) + w(S_{H \cup \{v\}})$. This ends the proof of this claim. $\diamondsuit$

Finally, when processing block $B_r$, the graph $G_{r-1}$ is completely reduced to a 1-split graph, so we can also apply the algorithm of Theorem 4 in order to obtain an optimal partition for $G_{r-1}$, and then recursively construct the solution for the graph $G_0 = G$, following the ideas in the Claim.

If the graph $G$ is unweighted, then the 1-split graphs in which we need to solve the subproblems have weight 1 in every internal edge. In that case, as we noticed in Theorem 4, the algorithm is very simple. $\square$

## 5 Interval graphs

Other interesting subclasses of chordal graphs are interval graphs. A graph $G$ is an *interval graph* if $G$ is the intersection graph of a finite family of intervals of the real line, and it is a *proper interval graph* if it admits an intersection model in which no interval properly contains another. A *unit interval graph* is the intersection graph of a finite family of intervals of the real line, all of the same length. Proper interval graphs and unit interval graphs coincide, and they are exactly the claw-free interval graphs [13] (the *claw* is the complete bipartite graph $K_{1,3}$).

A restricted subclass of unit interval graphs is the class of *paths of cliques*. A graph is a *path of cliques* if after identifying true twins into a single vertex, the resulting graph is a single path. That is, its vertex set can be partitioned into sets $A_1, \ldots, A_n$ in such a way that for each pair of vertices $v$, $w$, such that $v \in A_i$ and $w \in A_j$, they are adjacent if and only if either $i = j$ or $i = j + 1$ or $i = j - 1$. They also appear in the literature as *line graphs of multipaths*.

We will prove the next result with an approach similar to the proof od Theorem 4. We will define an initial solution and then we will code every other solution by a subset of a set. Finally, we will show that the function that assigns to every subset the difference of weights between its associated solution and the initial one is submodular.

**Theorem 6** (Polynomiality on weighted paths of cliques)**.** *The cluster deletion problem is polynomial-time solvable for weighted paths of cliques, when the weights are nonnegative.*

*Proof.* Let $A_1, \ldots, A_n$ be the vertex set partition of a path of cliques $G$. Without loss of generality we can assume that $n$ is even, by adding if necessary a set $A_{n+1}$ with only one vertex adjacent to every vertex in $A_n$ with edges of weight zero. Let $n = 2r$ and define the initial solution as the cliques $A_{2k-1} \cup A_{2k}$ for $1 \leq k \leq r$. Every vertex of $A_i$, for $1 < i < n$, has two possibilities. Either it will be part of a clique contained in $A_{i-1} \cup A_i$ or it will be part of a clique contained in $A_i \cup A_{i+1}$. So, an arbitrary solution is completely defined by the subset $S$ of $V(G) \setminus (A_1 \cup A_n)$ that, from the initial solution, are moved into a clique contained in $A_{2j} \cup A_{2j+1}$ for some integer $1 \leq j \leq r - 1$. We will define $f(S)$ for a subset $S$ of $V(G) \setminus (A_1 \cup A_n)$ as the difference between the weight of the solution associated with $S$ and the weight of the initial solution. The optimal solution will be given by the subset $S$ that minimizes function $f$. We will show now that $S$ is a submodular function and so a minimizer can be found in strongly polynomial time.

The function $f$ can be expressed as $f(S) = \sum_{1 \leq k \leq r} f^k(S)$, where

$$f^k(S) = \sum_{\substack{v \in S \cap A_{2k} \\ u \in A_{2k-1}}} w(vu) + \sum_{\substack{v \in S \cap A_{2k-1} \\ u \in A_{2k}}} w(vu) + \sum_{\substack{v \in S \cap A_{2k-1} \\ u \in A_{2k-1} \setminus S}} w(vu) + \sum_{\substack{v \in S \cap A_{2k} \\ u \in A_{2k} \setminus S}} w(vu) - \sum_{\substack{v \in S \cap A_{2k} \\ u \in S \cap A_{2k+1}}} w(vu) - \sum_{\substack{v \in S \cap A_{2k-1} \\ u \in S \cap A_{2k}}} w(vu)$$

where in order to simplify the expression, we let $A_{2r+1} = \emptyset$. Notice that the last term in the sum avoids the double counting of the first two terms.

For each value of $k$, define $f_1^k(S), \ldots, f_6^k(S)$ as the six terms of $f^k(S)$. It is easy to see that for $i = 1, 2$ and $S, T$ subsets of $V(G) \setminus (A_1 \cup A_n)$, it holds $f_i^k(S) + f_i^k(T) = f_i^k(S \cup T) + f_i^k(S \cap T)$. We will prove that, for $i = 3, \ldots, 6$, $f_i^k(S) + f_i^k(T) \geq f_i^k(S \cup T) + f_i^k(S \cap T)$.

For $S, T$ subsets of $V(G) \setminus (A_1 \cup A_n)$, and by decomposing $S$ as $(S \setminus T) \cup (S \cap T)$ (resp. $T$ as $(T \setminus S) \cup (S \cap T)$); $V(G) \setminus S$ as $(T \setminus S) \cup (V(G) \setminus (S \cup T))$ (resp. $V(G) \setminus T$ as $(S \setminus T) \cup (V(G) \setminus (S \cup T))$); $S \cup T$ as $(S \setminus T) \cup (T \setminus S) \cup (S \cap T)$; and $V(G) \setminus (S \cap T)$ as $(T \setminus S) \cup (S \setminus T) \cup (V(G) \setminus (S \cup T))$, it can be see that

$$f_3^k(S) + f_3^k(T) - f_3^k(S \cup T) - f_3^k(S \cap T) = 2 \sum_{\substack{v \in (S \setminus T) \cap A_{2k-1} \\ u \in (T \setminus S) \cap A_{2k-1}}} w(vu) \geq 0$$

because the weights are nonnegative, so $f_3^k$ is submodular. The proof for $f_4^k$ is identical.

Recall that $f_5^k(S) = -\sum_{v \in S \cap A_{2k}, u \in S \cap A_{2k+1}} w(vu)$. By using again the decomposition $S$ as $(S \setminus T) \cup (S \cap T)$ (resp. $T$ as $(T \setminus S) \cup (S \cap T)$); and $S \cup T$ as $(S \setminus T) \cup (T \setminus S) \cup (S \cap T)$, it can be see that

$$f_5^k(S) + f_5^k(T) - f_5^k(S \cup T) - f_5^k(S \cap T) = \sum_{\substack{v \in (S \setminus T) \cap A_{2k} \\ u \in (T \setminus S) \cap A_{2k+1}}} w(vu) + \sum_{\substack{v \in (T \setminus S) \cap A_{2k} \\ u \in (S \setminus T) \cap A_{2k+1}}} w(vu) \geq 0$$

because the weights are nonnegative, so $f_5^k$ is submodular. The proof for $f_6^k$ is identical. Finally, the sum of submodular functions is submodular, and this completes the proof. □

A split graph is called *complete* if each vertex of the stable set is adjacent to all vertices of the clique. By slightly modifying the proof of Theorem 1, we can prove the following.

**Theorem 7** (NP-completeness on weighted complete split graphs)**.** *The cluster deletion problem is NP-complete for weighted complete split graphs, even if the weight of all the internal edges of the clique is* 1*.*

*Proof.* We will reduce once more time the X3C problem. Let $X = \{x_1, \ldots, x_{3q}\}$ and $C = \{c_1, \ldots, c_m\}$ be an instance of the X3C problem, where each element $c_i \in C$ is a 3-element subset of $X$, with $m \geq q \geq 1$. We want to know if there exists a subset $C' \subseteq C$ with size $q$ such that each element in $X$ belongs to exactly one of the elements in $C'$. We will construct an edge-weighted complete split graph $G = (K_X \cup I_C, E)$, where $K_X$ induces a clique with $3q$ vertices and $I_C$ induces an independent set with $m$ vertices. In fact, for each element $x_i \in X$ there is a vertex $v_{x_i}$ in $K_X$ and for each 3-subset $c_j \in C$ there is a vertex $v_{c_j} \in I_C$. The edge set $E$ is formed as follows: $E = E_X \cup E_C \cup E_{\bar{C}}$, where $E_X = \{v_{x_i}v_{x_j} : i \neq j, v_{x_i}, v_{x_j} \in K_X\}$, $E_C = \{v_{c_j}v_{x_{j_1}}, v_{c_j}v_{x_{j_2}}, v_{c_j}v_{x_{j_3}} : v_{c_j} \in I_C, v_{x_{j_1}}, v_{x_{j_2}}, v_{x_{j_3}} \in K_X, c_j \in C \text{ and } c_j = \{x_{j_1}, x_{j_2}, x_{j_3}\}\}$, and $E_{\bar{C}} = \{v_{c_j}v_{x_i} : v_{c_j} \in I_C, v_{x_i} \in K_X, i \notin \{j_1, j_2, j_3\}, c_j \in C \text{ and } c_j = \{x_{j_1}, x_{j_2}, x_{j_3}\}\}$. Finally, the weight of each edge in $E_X \cup E_{\bar{C}}$ is equal to 1, and the weight of each edge in $E_C$ is equal to $\beta = \binom{3q}{2} + 3m(q-1) + 1$. Clearly, $G$ is a complete split graph and its construction is done in polynomial time. Let $W = \binom{3q}{2} - 3q + 3(m-q)\beta + 3m(q-1)$. We will show that there exists a subset $C' \subseteq C$, with $|C'| = q$, exactly covering $X$ if and only if $G$ has a clique partition where the sum of the weights of edges outside the cliques is at most $W$. In other words, there exists a solution for the X3C problem if and only if there exists an optimal solution for the cluster deletion problem for $G$ with weight at most $W$.

Assume first that there exists $C' \subseteq C$, with $C' = \{c'_1, \ldots, c'_q\}$ such that $c'_i \cap c'_j = \emptyset$ whenever $i \neq j$, and $\bigcup_{c'_j \in C'} c'_j = X$. The clique partition for $G$ can be constructed as follows: for each $c'_j \in C'$, with $c'_j = \{x_{j_1}, x_{j_2}, x_{j_3}\}$, choose the clique $\{v_{c'_j}, v_{x_{j_1}}, v_{x_{j_2}}, v_{x_{j_3}}\}$ in $G$. Each one of the remaining $m - q$ vertices in $I_C$ form a clique with size one. It is easy to see that the sum of the edge weights outside those cliques is exactly equal to $W$.

Conversely, assume that $G$ has a clique partition with weight at most $W$. We should to prove that there exists $C' \subseteq C$, with $|C'| = q$, such that $C'$ is an exact cover for $X$. For this, we analyze first the structure of such a solution. Notice that $\beta$ is greater than the sum of the weights of all the edges in $E_X \cup E_{\bar{C}}$, so a solution with weight $W$ should delete exactly $3(m-q)$ edges of $E_C$. It cannot delete less than $3(m-q)$ because each vertex of $K_X$ can be in the same clique with at most one vertex of $I_C$. So, every vertex of $K_X$ is in a clique with one vertex of $I_C$, and joined to that vertex by an edge of the set $E_C$. Than means that all the edges of $E_{\bar{C}}$ are deleted and that each clique contains at most three vertices of $K_X$. By the value of $W$, we can see that each clique should contain exactly three vertices of $K_X$, and following the lines of the proof of Theorem 1, such a solution corresponds to a solution of the X3C instance, which ends the proof of this theorem. □

Complete split graphs are also interval graphs, but (in general) they are not unit interval graphs, as if the stable set is of size at least three and the complete set is nonempty, they contain a *claw* (the complete bipartite graph $K_{1,3}$), that is not a unit interval graph. So, we have the following corollary.

**Corollary 1** (NP-completeness on weighted interval graphs)**.** *The cluster deletion problem is NP-complete for weighted interval graphs.*

For unweighted unit interval graphs instead, we will show that the cluster deletion problem is polynomial-time solvable. First, we need to state some known results and show a lemma describing the structure of an optimal solution.

**Theorem 8** (Roberts, 1969 [13])**.** *A graph $G$ is a unit interval graph if and only if its vertices can be linearly ordered such that, for each clique $M$ of $G$, the vertices contained in $M$ are consecutive.*

Such an ordering is called a *canonical ordering* of the vertices.

**Lemma 2** (Consecutiveness for unweighted unit interval graphs)**.** *Let $G$ be an unweighted unit interval graph and $v_1, \ldots, v_n$ be a canonical ordering of the vertices of $G$. Then there is an optimal solution of the cluster deletion problem for $G$, such that each clique of the solution consists of consecutive vertices in that ordering.*

*Proof.* Let us define, for each clique $B$ of the optimal solution $S$, $m(B) = \max\{j : v_j \in B\}$. Now, let $B_1, \ldots, B_k$ be the cliques of the solution ordered by $m$ increasingly. Suppose that not all the cliques consist of consecutive vertices, and let $i$ minimum such that either $i < m(B_1)$ and $v_i \notin B_1$, or $m(B_{j-1}) < i < m(B_j)$ but $v_i \notin B_j$, for some $j$. Let $j'$ such that $v_i \in B_{j'}$. Then, by the choice of $i$, all the vertices of $B_{j'}$ have subindex greater than $i$ and $j' > j$ so $m(B_{j'}) > m(B_j)$. Since $v_i$ is adjacent to $v_{m(B_{j'})}$ and $G$ is a proper interval graph with canonical ordering $v_1, \ldots, v_n$, the vertices $v_i, \ldots, v_{m(B_{j'})}$ form a clique and, in particular, $B_{j'} \cup \{v_{m(B_j)}\}$ is a clique and $v_i$ is adjacent to $v_{m(B_j)}$. So, independently of $i$ being greater os less than the minimum index of a vertex in $B_j$, $B_j \cup \{v_i\}$ is a clique. Then, either $|B_j| \geq |B'_j|$ and $S \setminus \{B_j, B_{j'}\} \cup \{B_j \cup \{v_i\}, B_{j'} \setminus \{v_i\}\}$ is a solution strictly better than $S$, or $|B_j| < |B'_j|$ and $S \setminus \{B_j, B_{j'}\} \cup \{B_j \setminus \{v_{m(B_j)}\}, B_{j'} \cup \{v_{m(B_j)}\}\}$ is a solution strictly better than $S$, a contradiction in both cases. □

**Theorem 9** (Polynomiality on unweighted unit interval graphs)**.** *The cluster deletion problem can be solved in polynomial time on unweighted unit interval graphs.*

*Proof.* By using Lemma 2, we can easily develop a dynamic programming algorithm. For $i = 0, 1, \ldots, n$, let $f(i)$ be the value of an optimal cluster deletion solution for the subgraph of $G$ induced by $v_1, \ldots, v_i$. Then $f(0) = f(1) = 0$ and, for $i > 1$, $f(i)$ will be the minimum over all the possible cliques containing $v_i$, i.e., $\{v_j, \ldots, v_i\}$ where either $j = i$ or $v_j v_i \in E(G)$, of $f(j-1)$ plus the amount of vertices joining $\{v_1, \ldots, v_{j-1}\}$ with $\{v_j, \ldots, v_i\}$ (this is 0 if $j = 1$). By keeping also the number $j$ realizing the minimum $f(i)$, we can also reconstruct the partition itself. □

General interval graphs do not have the same clique structure as unit interval graphs. And for weighted unit interval graphs, Lemma 2 does not hold. An example of this is the graph $P_6^2$, whose vertices are $v_1, \ldots, v_6$ and $v_i$ is adjacent to $v_j$ if and only if $1 \leq |i - j| \leq 2$. It is easy to see that the only possible canonical orderings for $P_6^2$ are $v_1, \ldots, v_6$ or $v_6, v_5, \ldots, v_1$. Let $w$ be defined on the edges of $P_6^2$ such that $w(v_2v_4) = w(v_3v_5) = 100$ and $w(e) = 1$ for every other edge $e$. Any solution to the cluster deletion problem that does not contain $\{v_2, v_4\}$ and $\{v_3, v_5\}$ as cliques has weight at least 100, so the optimal solution is to have $\{v_2, v_4\}$, $\{v_3, v_5\}$, and isolated vertices, with a weight equal to 7.

The example shows that the idea of Theorem 9 cannot be generalized in a straightforward way. It remains as an open question the computational complexity of the cluster deletion problem on unweighted interval graphs and on weighted unit interval graphs.

Complete split graphs are also cographs (i.e. $P_4$-free graphs, that are not necessarily chordal as they may contain $C_4$ as induced subgraph). The cluster deletion problem on unweighted cographs was solved in polynomial time by Gao et al. in [9]. As a corollary of Theorem 7, we have this complexity result for the weighted case.

**Corollary 2** (NP-completeness on weighted cographs)**.** *The cluster deletion problem is NP-complete for weighted cographs.*

## 6 Further results and open problems

We showed in Theorem 1 that the cluster deletion problem is NP-complete for weighted 3-split graphs even if the weight of all the internal edges of the clique is 1 and the weight of the edges between the clique and the stable set is uniform. We have seen also in Theorem 4 that the cluster deletion problem is polynomial-time solvable for weighted 1-split graphs. As for 2-split graphs, we will show next that, under the conditions of Theorem 1 (the weight of all the internal edges of the clique is 1 and the weight of the edges between the clique and the stable set is uniform) the problem is polynomial-time solvable.

**Theorem** (Polynomiality on restricted weighted 2-split graphs)**.** *The cluster deletion problem is polynomial-time solvable for weighted* 2*-split graphs, if the weight of all the internal edges of the clique is* 1 *and the weight of the edges between the clique and the stable set is uniform.*

*Proof.* Let $G$ be a 2-split graph with split partition $(K, I)$, and $\beta$ be the weight of the edges between $K$ and $I$. Let us create a graph $G'$ with vertex set $K'$, where $K'$ is the subset of vertices of $K$ that have at least one neighbor in $I$, and such that two vertices are adjacent in $G'$ if they have a common neighbor of $I$ in $G$. Minimizing the sum of the external edges of a clique partition of $G$ is equivalent to maximizing the sum of the internal edges of the clique partition. There are three possible class of cliques in $G$: those containing one vertex of $I$ and two vertices of $K$, those containing one vertex of $I$ and one vertex of $K$, or those completely included in $K$, and by optimality, there is

just one of such cliques. Then the sum of internal edges will be $a(2\beta+1)+b\beta+c(c-1)/2$ $(*)$, where $0 \leq a \leq \nu(G')$ ($\nu(G')$ is the value of a maximum matching of $G'$), $0 \leq b \leq |K'|-2a$, and $c = |K|-2a-b$. It is easy to see that in an optimal solution, either $a = \nu(G')$ or $b = 0$. In the first case, after the substitution $a = \nu(G')$, the coefficient of $b$ in the expression $(*)$ is positive, so the maximum is attained either by $b = 0$ or by $b = |K'| - 2\nu(G')$. In the second case, after the substitution $b = 0$, the coefficient of $a$ in the expression $(*)$ is positive, so the maximum is attained either by $a = 0$ or by $a = \nu(G')$. So the problem is reduced to solve maximum matching in $G'$ and then compute the value of $(*)$ for the three possible optimal solutions $a = 0, b = 0$; $a = \nu(G'), b = 0$; and $a = \nu(G'), b = |K'| - 2\nu(G')$. □

We leave as an open problem the computational complexity of the cluster deletion problem in general weighted 2-split graphs, or even when the weight of all the internal edges of the clique is 1 but the weight of the edges between the clique and the stable set is arbitrary and not necessarily uniform.

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